



**Rajshahi University of Engineering & Technology**

Department of Mechatronics Engineering

# LAB REPORTS

Course no: **MTE 3206**

Course Title: **Hydraulic and Pneumatic Control Sessional**

Experiment No: **02**

Name of the Experiment: Study of FluidSIM Software and Investigation of Basic Operations.

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## **Experiment No.: 02**

**Experiment Name:** Study of FluidSIM Software and Investigation of Basic Operations

### **Objectives**

- Understand the functionality of FluidSIM software.
- Explore basic pneumatic system operations using FluidSIM.
- Design and simulate logic gates with pneumatic components in FluidSIM.

### **Theory**

FluidSIM, developed by Festo, is a leading simulation tool for designing pneumatic, hydraulic, and electrical systems. It supports circuit creation, real-time simulation, and performance optimization. FluidSIM is widely used in teaching automation technologies, allowing hands-on simulation and interactive learning [1].

### **Key Features**

- Interactive Simulation: Real-time behavior analysis of circuits.
- Rich Component Libraries: Extensive pneumatic, hydraulic, and electrical elements.
- Measurement Tools: Real and virtual instruments for system monitoring.
- Fault Simulation: Diagnose and troubleshoot system errors.
- Flexible Access: Available both online and offline for broad usability.

### **Required Apparatus**

- FluidSIM software
- A computer or compatible device
- Pneumatic design reference materials

### **Operations**

#### **3/2 Directional Control Valve**

A 3/2 directional control valve features three ports and two positions, making it well-suited for controlling single-acting cylinders. The ports consist of an air supply inlet, an outlet connected to the actuator, and an exhaust port for air release. In its default (non-actuated) position, the inlet is closed, and the outlet is linked to the exhaust, allowing the actuator to reset. When actuated, the inlet opens to the outlet, directing compressed air to the actuator to initiate movement. These valves are highly adaptable and can be operated manually, mechanically, or electrically, depending on system needs. They are commonly used in pneumatic systems requiring straightforward on-off airflow control.

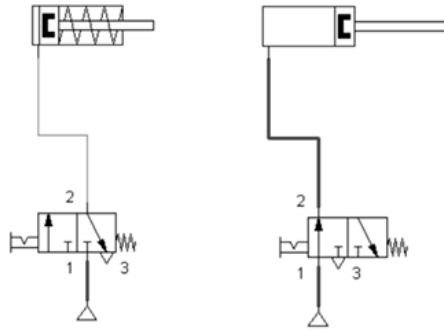


Fig.1. 3/2 Directional Control Valve in non-actuated and actuated state

### 5/2 Directional Control Valve:

A 5/2 directional control valve is designed with five ports and two operating positions, making it perfect for managing double-acting cylinders. The ports consist of an air inlet, two outlets that connect to opposite ends of the cylinder, and two exhaust ports to release air from the inactive side. In one position, air is directed to one outlet while the other outlet is linked to its exhaust, causing the cylinder to move in one direction. When the valve switches positions, the airflow reverses, moving the cylinder in the opposite direction. These valves are typically activated by solenoids, pilot air signals, or mechanical means, ensuring accurate control and smooth operation for applications requiring back-and-forth motion.

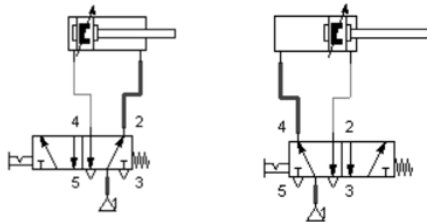


Fig.2. 5/2 Directional Control Valve in non-actuated and actuated state

### Normally Open and Normally Closed Valves:

- Normally Open Valve:** In the default state, the valve allows air to flow freely from input (0) to output (1). When actuated, it switches, blocking the airflow from input (1) to output (0). This function is typically achieved using a 3/2 valve. In its normal state, the valve provides an unrestricted air supply. However, once activated, it interrupts the flow of air into the cylinder. These valves are commonly found in fail-safe systems where the flow of air needs to be cut off upon activation to ensure safety.
- Normally Closed Valve:** This is also done using 3/2 valve. By default, this valve prevents air from flowing. It only opens when actuated, allowing air to pass through. These valves are suitable for systems where airflow should only occur during operation or on-demand control [2].

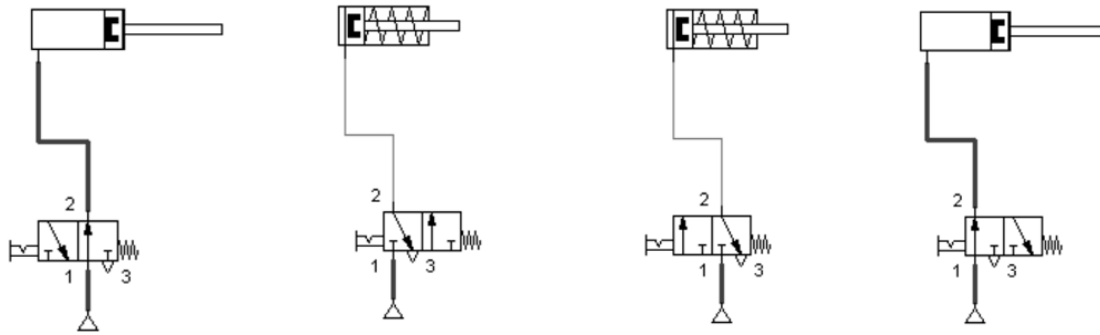


Fig.3. Normally open and Normally close valve in non-actuated and actuated states

### External Mass

When an external force of up to 500 kg is applied, the cylinder moves at a slower pace due to the Additional load. However, the operation of the system remains unchanged, maintaining its functionality as it did prior to the external mass being applied.

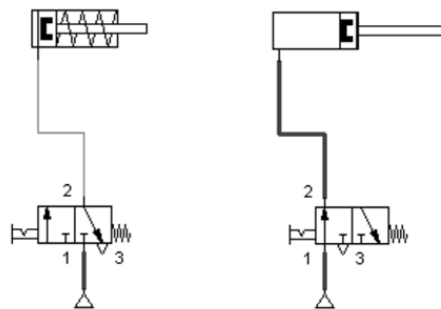


Fig.4. 3/2 directional control valve in non-actuated and actuated states with 500kg external load applied

### Flow and Pressure Control Valves:

- Non-Return Valve:** Also called a check valve, a non-return valve ensures that air flows in only one direction, preventing backflow in pneumatic circuits. This valve is essential for maintaining pressure and preventing disruption caused by reverse airflow. It is commonly used in systems where unidirectional airflow is necessary, such as in maintaining pressure in a compressed air reservoir or in circuits with multiple actuators.

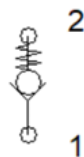


Fig.5. Non-Return Valve or Check Valve

- **Throttle Valve:** A throttle valve regulates the rate of airflow in a pneumatic system, directly affecting the speed of actuators like cylinders. By reducing the available cross-sectional area for air passage, it limits the flow rate. These valves are adjustable, providing precise control over the speed and smoothness of cylinder movement in both the extending and retracting directions. Throttle valves are crucial for applications where controlling motion speed and smoothness is important.



Fig.6. Throttle Valve

- **Unidirectional Flow Control Valve:** A unidirectional flow control valve combines a throttle valve and a non-return valve, allowing controlled airflow in one direction while permitting unrestricted flow in the opposite direction. This dual functionality allows for precise control of cylinder speed during either the outstroke or instroke, depending on the valve's installation. It's ideal for circuits requiring smooth movement in one direction and free airflow in the other.

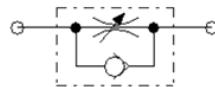


Fig.7. Unidirectional Flow Control Valve

- **Shuttle Valve:** Also known as an OR valve, a shuttle valve directs air from two separate inputs to a single output. It contains a shuttle mechanism (usually a small ball) that blocks one inlet while allowing air from the other to flow through to the output. Shuttle valves are commonly used in circuits that require alternate control from two inputs, such as manual and automatic operations, ensuring that only one signal reaches the actuator at a time.

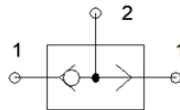


Fig.8. Shuttle Valve

- **Two-Pressure Valve:** Also called an AND valve, a two-pressure valve requires simultaneous inputs from two independent sources to produce an output. This design ensures that both conditions must be satisfied before the actuator operates, making it ideal for safety systems. For example, it can be used to confirm both the presence of an operator and the machine guard before activation, preventing accidental operation [3].

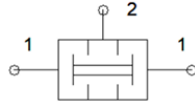


Fig.9. Two Pressure Valve

### Logic Gates in Pneumatic Systems:

In pneumatic systems, logic gates can be represented using air flow and control valves, where air flow behaves similarly to electrical signals:

- High pressure (air flow) represents a logical 1.
- Low pressure (no air flow) represents a logical 0.

### AND Gate (Series Connection)

- **Pneumatic Representation:** In this configuration, two 3/2 valves are connected in series. Air can flow through the system only when both valves are open, allowing airflow.
- **Logical Behavior:** The output (air flow) is 1 (air flows) only when both inputs are 1 (both valves are open). If either or both valves are closed (inputs are 0), no air flows.

| Input A | Input B | Output |
|---------|---------|--------|
| 0       | 0       | 0      |
| 0       | 1       | 0      |
| 1       | 0       | 0      |
| 1       | 1       | 1      |

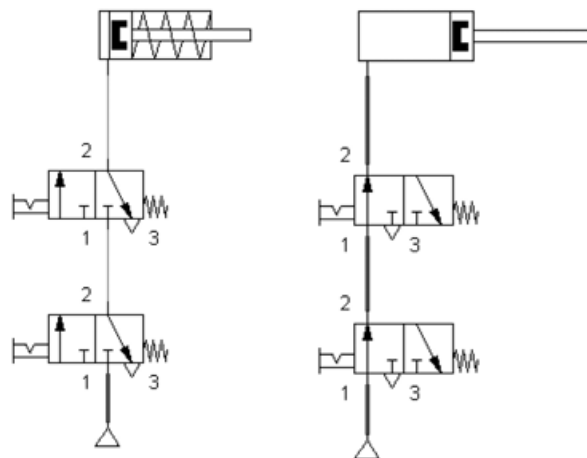


Fig.10. AND Gate Series Connection in (0,0) & (1,1) states

## OR Gate (Parallel Connection)

- **Pneumatic Representation:** In this setup, two valves are connected in parallel using a Shuttle Valve. Air can flow if either or both valves are open.
- **Logical Behavior:** The output (air flow) is 1 (air flows) if at least one input is 1 (one or both valves are open). If both valves are closed (inputs are 0), no air flows.

| Input A | Input B | Output |
|---------|---------|--------|
| 0       | 0       | 0      |
| 0       | 1       | 1      |
| 1       | 0       | 1      |
| 1       | 1       | 1      |

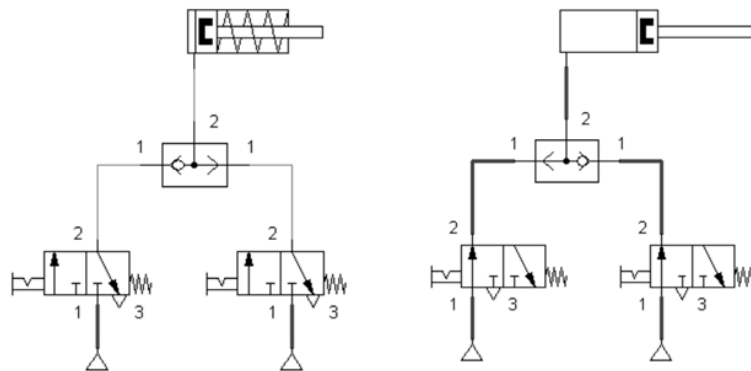


Fig.11. OR Gate Parallel Connection in (0,0) & (1,1) states

## NOT Gate (Inverter)

- **Pneumatic Representation:** A valve is designed to invert the signal. When the switch is on, the airflow is blocked, and the cylinder moves to the negative position (instroke). In the initial state, air flows into the cylinder, and the piston moves to the positive position (outstroke).
- **Logical Behavior:** The output is 1 (air flows) when the input is 0 (no air flows). If the input is 1 (air flows), the output is 0 (air is blocked).

| Input | Output |
|-------|--------|
| 0     | 1      |
| 1     | 0      |

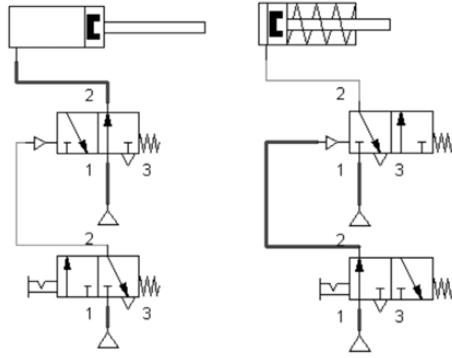


Fig.12. NOT Gate in Switch OFF and ON states

## NAND Gate

- **Pneumatic Representation:** A NAND gate can be created by combining an OR gate using a shuttle valve and Normally Open 3/2 valves. Air flow is blocked only when both switches are pressed (i.e., both valves are open).
- **Logical Behavior:** The output is 1 unless both inputs are 1 (when both valves are open). If both inputs are 1, the NOT gate blocks the airflow, resulting in a 0 output.

| Input A | Input B | Output |
|---------|---------|--------|
| 0       | 0       | 1      |
| 0       | 1       | 1      |
| 1       | 0       | 1      |
| 1       | 1       | 0      |

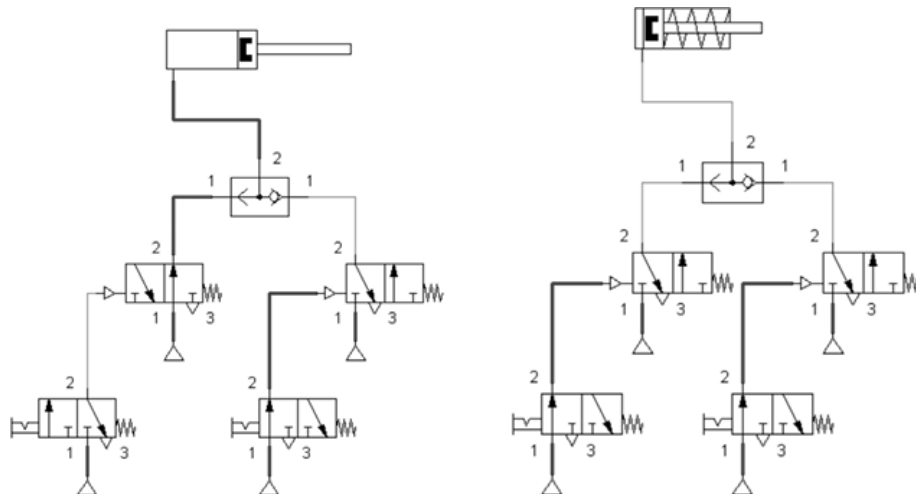


Fig.13. NAND Gate in (0,1) & (1,1) states

## NOR Gate

- **Pneumatic Representation:** A NOR gate is created by combining an AND gate with a Two-Pressure valve and a Normally Closed 3/2 Directional Control Valve (DCV). Air flow occurs only when both switches are not pressed (both valves are closed).
- **Logical Behavior:** The output is 1 only when both inputs are 0 (both valves are closed). If either or both inputs are 1, the NOT gate blocks the airflow, resulting in a 0 output.

| Input A | Input B | Output |
|---------|---------|--------|
| 0       | 0       | 1      |
| 0       | 1       | 0      |
| 1       | 0       | 0      |
| 1       | 1       | 0      |

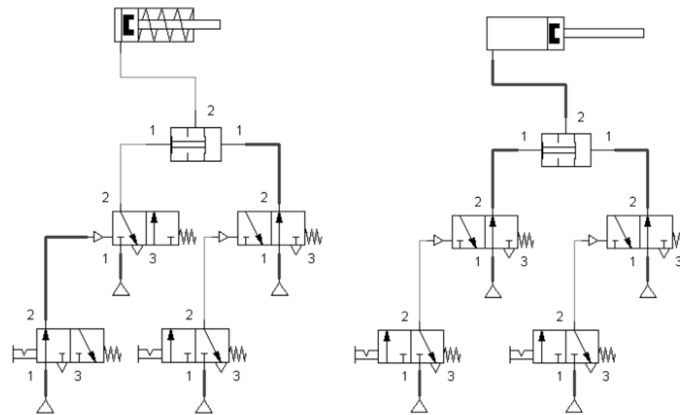


Fig.14. NOR Gate in (1,0) & (1,1) states

## XOR Gate (Exclusive OR)

- **Pneumatic Representation:** In an XOR gate, two valves are arranged so that air flows only when one valve is open, but not both. This can be achieved using a special valve design that directs air flow in divergent paths.
- **Logical Behavior:** The output is 1 only when the inputs are different (one valve open, the other closed). If both valves are open or both are closed, no air flows.

| Input A | Input B | Output |
|---------|---------|--------|
| 0       | 0       | 0      |
| 0       | 1       | 1      |
| 1       | 0       | 1      |
| 1       | 1       | 0      |

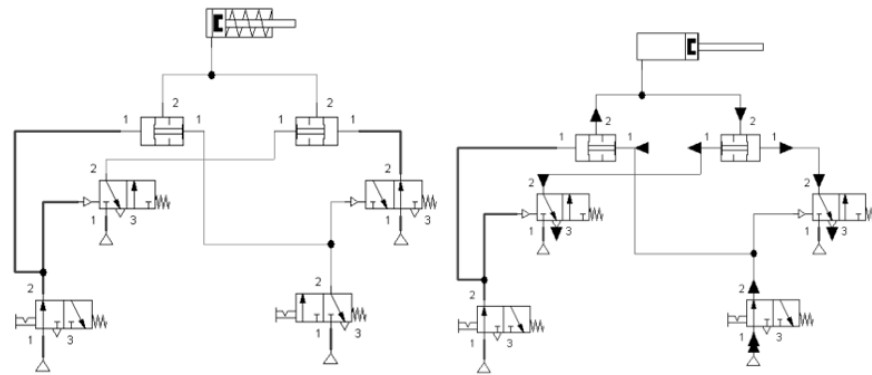


Fig.15. XNOR Gate in (1,0) & (1,1) states

## Discussion and Conclusion

In this experiment, we delved into the capabilities of FluidSIM, an advanced simulation tool for designing and testing pneumatic systems. Our exploration focused on the functionality of key components, such as 3/2 and 5/2 directional control valves, normally open and closed valves, and their role in controlling both single-acting and double-acting cylinders. We also investigated various flow and pressure control devices, including non-return, throttle, unidirectional flow control, shuttle, and two-pressure valves. These components are crucial for regulating airflow and ensuring the precision and reliability of pneumatic systems. A significant portion of the study was dedicated to designing and simulating logic gates (AND, OR, NOT, NAND, NOR, XOR, and XNOR) using pneumatic components. This demonstrated how pneumatic systems can integrate logical functions, providing a real-world application for these gates in automation. The ability to visualize and simulate such complex circuits allowed us to better understand how different pneumatic components work together to achieve desired outcomes.

This hands-on investigation not only reinforced theoretical concepts learned in the classroom but also highlighted the versatility of FluidSIM. By bridging theoretical knowledge with practical simulation, FluidSIM provides an excellent educational tool, enhancing our understanding of automation technology in a dynamic and user-friendly environment. Through this experiment, we gained deeper insights into the design, control, and optimization of pneumatic systems, reinforcing their relevance in modern automation applications.

## References

- [1] Festo.com. [Online]. Available: [https://www.festo.com/us/en/e/technical education/digital-learning/virtual-simulation-and-modeling/fluidsim-id\\_1663056/](https://www.festo.com/us/en/e/technical%20education/digital-learning/virtual-simulation-and-modeling/fluidsim-id_1663056/). [Accessed: 16-Jan-2025].
- [2] Festo.com. [Online]. Available: [https://www.festo.com/gb/en/e/journal/in practice/pneumatic valves](https://www.festo.com/gb/en/e/journal/in%20practice/pneumatic%20valves). [Accessed: 16 Jan-2025].
- [3] P. By, "Shuttle valve & Two pressure valve," Weebly.com. [Online]. Available: [https://maysaaiaat.weebly.com/uploads/5/8/8/3/5883161/atm1122\\_pneumatics\\_module\\_student\\_version.pdf](https://maysaaiaat.weebly.com/uploads/5/8/8/3/5883161/atm1122_pneumatics_module_student_version.pdf). [Accessed: 16-Jan-2025].